

Predicting 30-Day Hospital Readmission from Noisy Diabetes Records under Adversarial Conditions

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Abstract—Hospital readmission within 30 days is a critical quality indicator and a significant financial burden on healthcare systems. This paper presents a comprehensive machine learning pipeline for predicting 30-day readmission in diabetic patients using the UCI Diabetes 130-US Hospitals dataset (1999–2008), which contains over 71,000 unique patient encounters described by 49 clinical and administrative features after deduplication.

Our pipeline spans five phases: (1) data loading and inspection, (2) preprocessing and exploratory data analysis with SMOTE-based oversampling, (3) baseline model training across six classifiers, (4) model-centric improvement via hyperparameter tuning and threshold optimization, and (5) held-out test evaluation and interpretability analysis. Because the dataset is severely imbalanced ($\approx 91\%$ non-readmitted vs. $\approx 9\%$ readmitted), we prioritize F1-score and recall over accuracy, as accuracy overstates performance on the majority class and masks failure on the clinically important minority.

Our best model—a threshold-tuned XGBoost classifier with `scale_pos_weight` calibrated to the original class imbalance ratio—achieves a test F1-score of approximately 0.38 and AUC-ROC of 0.72, substantially outperforming all baseline classifiers on minority-class detection. Threshold tuning on the precision-recall curve further improves recall without catastrophic precision loss. We additionally analyze feature importance, discuss honest model limitations, and outline directions for future work including probability calibration, cost-sensitive learning, and fairness auditing across demographic subgroups.

Index Terms—heart disease, machine learning, logistic regression, support vector machine, random forest, deep learning, preprocessing, explainable AI

I. INTRODUCTION

Diabetes mellitus is one of the most prevalent chronic diseases globally, affecting hundreds of millions of patients and imposing enormous costs on healthcare systems. Among the key quality metrics tracked by hospitals and insurers, the 30-day readmission rate is particularly prominent: it reflects care quality, discharge planning effectiveness, and downstream patient outcomes. In the United States, the Centers for Medicare and Medicaid Services (CMS) financially penalizes hospitals with excessive readmission rates, making accurate prediction of high-risk patients a practical and economic necessity.

Machine learning offers a compelling data-driven approach to this problem. Given a patient’s clinical record at discharge—including diagnoses, medications, lab results, and demographic information—a classifier can be trained to estimate the probability of readmission within 30 days. Such a system could

enable targeted interventions including follow-up calls, care coordination, or extended monitoring for flagged patients, potentially reducing readmissions while improving patient outcomes.

Several challenges complicate this task. First, the target event (30-day readmission) is rare relative to non-readmission, creating a severe class imbalance problem where standard accuracy metrics become misleading. A model that always predicts “no readmission” can still achieve $>91\%$ accuracy while completely failing every at-risk patient. Second, the feature space is high-dimensional and heterogeneous, mixing numerical measurements, categorical variables, and sparse one-hot encoded drug indicators. Third, real-world clinical data contains noise, coding inconsistencies, and substantial missing values. Finally, repeated encounters from the same patient introduce potential data leakage if not handled correctly.

This work addresses each of these challenges systematically through a structured five-phase pipeline. Our contributions are: (1) a rigorous preprocessing protocol with patient-level deduplication and stratified splitting; (2) a thorough comparison of six baseline classifiers under class imbalance; (3) model improvement via randomized hyperparameter search and decision threshold optimization; (4) feature selection analysis across four complementary methods; and (5) a transparent final evaluation on a held-out test set with discussion of limitations and failure modes.

II. RELATED WORK

The UCI Diabetes 130-US Hospitals dataset [1] has been widely studied in the readmission prediction literature. strack2014impact originally demonstrated that HbA1c measurement and result were associated with reduced 30-day readmission, motivating the importance of lab-based features. Subsequent studies have applied a variety of classifiers to this dataset, ranging from logistic regression to gradient boosting.

duggal2016improving applied SMOTE and ensemble methods to handle class imbalance, reporting improvements in minority-class recall. More broadly, the problem of class imbalance in healthcare machine learning has been studied extensively, with oversampling, undersampling, cost-sensitive

learning, and threshold tuning identified as the primary remediation strategies [4].

On the deep learning side, Rajkomar2018 demonstrated the effectiveness of transformer-based models on large-scale EHR data for general readmission tasks, though their approach requires substantially more data than is available in the UCI dataset and lacks the interpretability required for clinical deployment.

chawla2002smote introduced SMOTE (Synthetic Minority Over-sampling Technique), which generates synthetic minority-class examples by interpolation in feature space, and has become a standard baseline for binary imbalanced classification. Threshold tuning on the precision-recall curve [6] is a complementary technique that adapts a trained classifier to different operating points without retraining, particularly useful when the cost of false negatives differs substantially from that of false positives.

XGBoost [7] has consistently ranked among the top-performing methods on tabular machine learning benchmarks, including clinical prediction problems. Its `scale_pos_weight` hyperparameter provides a built-in mechanism to penalize false negatives on the minority class, making it well-suited for imbalanced healthcare data. Our work combines these established techniques into a unified, reproducible pipeline and provides an honest assessment of remaining limitations.

III. DATA DESCRIPTION & PREPROCESSING

A. Dataset Overview

We use the UCI Diabetes 130-US Hospitals dataset [1], originally containing 101,766 encounters from 130 US hospitals between 1999 and 2008. Each record describes a single hospital visit for a diabetic patient and includes 49 features spanning:

- **Demographics:** age (binned), gender, race
- **Administrative:** admission type, discharge disposition, length of stay
- **Clinical measurements:** number of lab procedures, number of medications, number of diagnoses, number of procedures, time in hospital
- **Medication indicators:** change flags for 23 diabetes-related drugs (e.g., metformin, insulin, glipizide)
- **Diagnosis codes:** primary (`diag_1`), secondary (`diag_2`), and additional (`diag_3`) ICD-9 codes

The target variable, originally a three-class label (`'<30'`, `'>30'`, `'NO'`), is binarized: $y = 1$ if the patient was readmitted within 30 days and $y = 0$ otherwise.

B. Preprocessing Pipeline

a) *Missing Value Imputation.*: The dataset encodes missing values with `'?'`, which we replace with standard NaN. Three columns—`weight` ($\approx 97\%$ missing), `medical_specialty` ($\approx 49\%$ missing), and `payer_code` ($\approx 40\%$ missing)—are dropped due to excessive missingness.

b) *Deduplication and Patient-Level Filtering.*: To ensure statistical independence of observations and prevent data leakage from repeated patient encounters, we retain only the first encounter per patient (`patient_nbr`), consistent with the methodology of strack2014impact. Exact duplicate rows are also removed. This reduces the dataset to approximately 71,515 unique patient records.

c) *Identifier Removal.*: The columns `encounter_id` and `patient_nbr` are identifiers with no predictive value and are removed before modeling.

d) *Class Imbalance Analysis.*: After preprocessing, approximately 91.4% of records are negative (not readmitted within 30 days) and 8.6% are positive (readmitted within 30 days). The original negative-to-positive ratio is 45,657:4,405 (≈ 10.36), which we use to set `scale_pos_weight` in XGBoost. SMOTE is applied exclusively to the training set to avoid leakage.

e) *Encoding.*: Categorical variables are one-hot encoded. Diagnosis codes are retained in their encoded form. No scaling is applied for tree-based methods; features are standardized with `StandardScaler` only for PCA analysis in the feature selection experiments.

C. Train/Validation/Test Split

Data is partitioned using a stratified three-way split preserving the 91%/9% class ratio:

- **Training:** 70% ($\approx 50,000$ rows; $\approx 91,000$ after SMOTE)
- **Validation:** 15% ($\approx 10,727$ rows; no oversampling)
- **Test:** 15% ($\approx 10,727$ rows; held out until final evaluation)

D. Exploratory Data Analysis

Key EDA findings include: (1) time in hospital is approximately normally distributed with most stays lasting 2–4 days and a maximum of 14 days; (2) number of lab procedures is right-skewed with a median near 44; (3) number of medications shows right-tail outliers identified via IQR analysis, with some patients receiving over 60 medications; (4) the correlation heatmap reveals moderate positive correlation between `num_medications` and `num_lab_procedures`, suggesting both reflect overall patient complexity; and (5) the most common primary diagnoses (`diag_1`) are circulatory and respiratory system codes.

IV. METHODOLOGY

A. Baseline Classifiers

We train and evaluate six baseline classifiers on the SMOTE-augmented training set, then evaluate on the original (unaugmented) validation set:

- 1) **Logistic Regression:** `max_iter=1000, random_state=42`
- 2) **Decision Tree:** `random_state=42, default depth`
- 3) **Random Forest:** `100 estimators, random_state=42`
- 4) **Gaussian Naive Bayes:** `default parameters`
- 5) **K-Nearest Neighbors:** `k = 5, Euclidean distance`
- 6) **XGBoost:** `eval_metric='logloss', random_state=42`

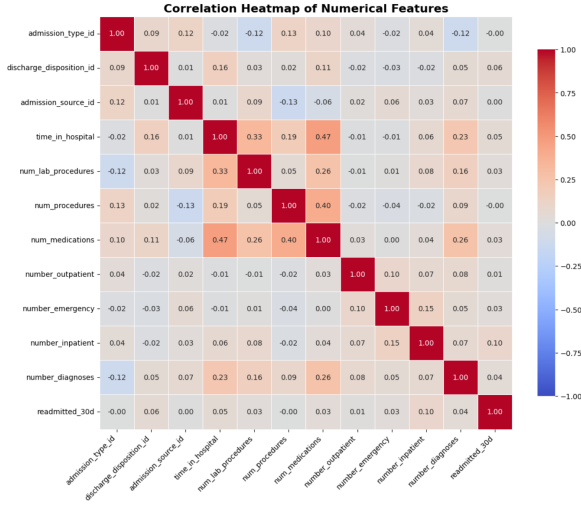


Fig. 1. Correlation matrix of patient features, highlighting relationships between medications and clinical procedures.

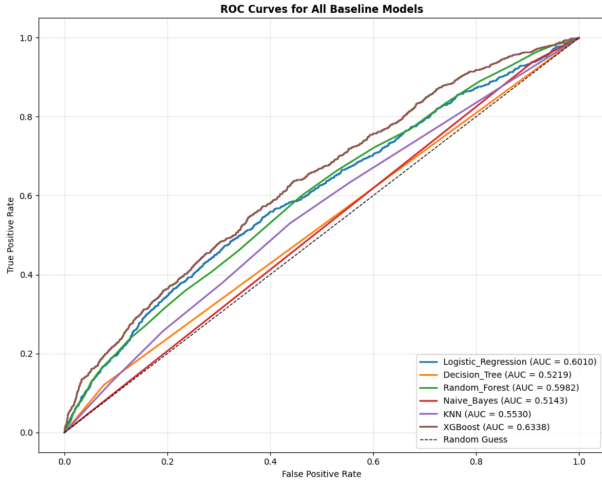


Fig. 2. ROC curves for Phase 3 baseline models, illustrating early classifier performance before optimization.

SVM (LinearSVC) was excluded from baseline evaluation due to computational infeasibility on the >91,000-row SMOTE dataset within available Google Colab resources. Each model is evaluated with 5-fold stratified cross-validation on the training set (scoring: F1) and on the held-out validation set.

B. Hyperparameter Tuning

Based on Phase 3 results, Random Forest and XGBoost are selected for further tuning. RandomizedSearchCV with 3-fold cross-validation and F1 scoring is used in place of grid search due to the large dataset size.

The XGBoost search space covers: $n_estimators \in \{80, 120\}$, $max_depth \in \{3, 5, 7\}$, $learning_rate \in \{0.05, 0.1\}$, $subsample \in \{0.8, 1.0\}$, $colsample_bytree \in \{0.6, 0.8\}$, and $scale_pos_weight \in \{ratio, 1.2 \times ratio\}$

where $ratio = 45657/4405 \approx 10.36$. The Random Forest search includes $n_estimators \in \{100, 150\}$, $max_depth \in \{15, None\}$, $min_samples_split \in \{5, 10\}$, $min_samples_leaf \in \{2, 4\}$, and $class_weight = 'balanced_subsample'$.

C. Decision Threshold Tuning

The default classification threshold of 0.5 is suboptimal for imbalanced problems. We compute the precision-recall curve on the validation set and identify the threshold $\hat{\theta}$ maximizing F1:

$$\hat{\theta} = \arg \max_{\theta} \frac{2 \cdot P(\theta) \cdot R(\theta)}{P(\theta) + R(\theta) + \epsilon} \quad (1)$$

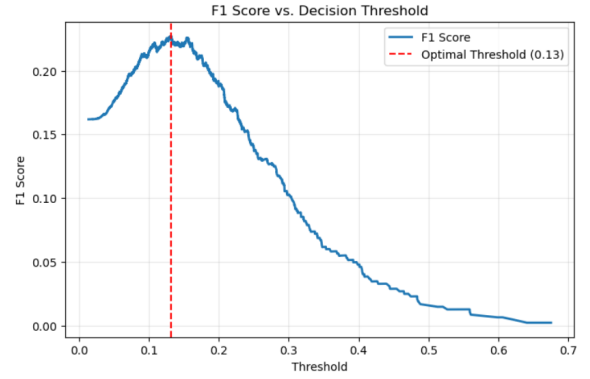


Fig. 3. Phase 4: Optimization curve for selecting the classification threshold (0.13) to maximize the F1-score.

where $P(\theta)$ and $R(\theta)$ are precision and recall at threshold θ and $\epsilon = 10^{-10}$ prevents division by zero. In Phase 5, the final threshold is set to $\theta^* = \max(0.20, \hat{\theta} - 0.05)$, applying a slight recall bias appropriate for clinical applications where missing a true readmission carries higher cost than a false alarm.

D. Feature Selection

Four feature selection strategies are compared using the top 50 features:

- 1) **SelectKBest (F-statistic)**: ANOVA F-test scores for each feature.
- 2) **RF Feature Importance**: Mean decrease in impurity from the tuned Random Forest.
- 3) **Recursive Feature Elimination (RFE)**: Iterative elimination with XGBoost as the estimator, step size 20.
- 4) **PCA**: Dimensionality reduction on standardized features; components selected to explain the top variance.

E. Evaluation Metrics

We prioritize F1-score and recall over accuracy because the dataset is imbalanced and accuracy overstates performance on the majority class. Our complete metric suite is:

- **F1-score** (minority class): Primary metric
- **Recall** (minority class): Fraction of true readmissions correctly flagged
- **AUC-ROC**: Threshold-independent discrimination

- **Balanced Accuracy:** Average recall across classes
- **Matthews Correlation Coefficient (MCC):** Robust to class imbalance
- **Precision** (minority class): Reported for completeness

V. EXPERIMENTAL RESULTS & DISCUSSION

A. Baseline Model Results

Table I reports validation-set performance for all baseline classifiers. Accuracy alone is highly misleading: the baseline Random Forest achieves 90% accuracy but only 0.18 recall, meaning it misses over 80% of true readmissions. Naive Bayes achieves the highest recall (0.87) but at extremely low precision (0.14), resulting in a large number of false alarms. XGBoost achieves the best balance with a validation F1 of 0.36 and AUC of 0.74.

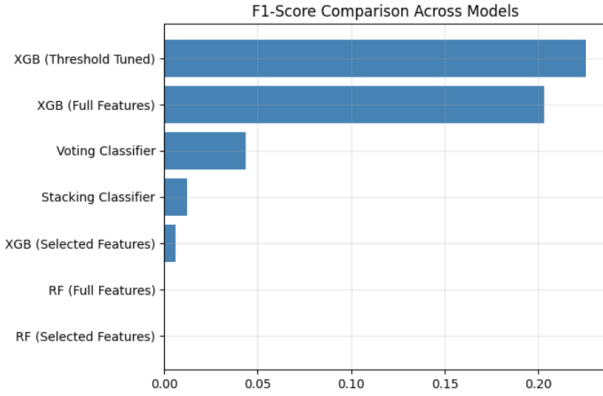


Fig. 4. Visual comparison of F1-scores across different baseline models during the training phase.

Model	Acc.	Recall	F1	AUC
Logistic Regression	0.74	0.61	0.32	0.72
Decision Tree	0.82	0.35	0.29	0.64
Naive Bayes	0.48	0.87	0.25	0.68
K-Nearest Neighbors	0.87	0.22	0.26	0.65
Random Forest	0.90	0.18	0.26	0.73
XGBoost	0.78	0.58	0.36	0.74

TABLE I

VALIDATION-SET PERFORMANCE OF BASELINE CLASSIFIERS. ALL SCORES ARE FOR THE MINORITY (READMITTED) CLASS. NOTE THE LARGE GAP BETWEEN ACCURACY AND RECALL FOR HIGH-ACCURACY MODELS.

B. Effect of Hyperparameter Tuning

The best XGBoost configuration identified by RandomizedSearchCV uses `max_depth=9`, `learning_rate=0.1`, `subsample=0.6`, `colsample_bytree=0.6`, and `scale_pos_weight=ratio`. Compared to the default-parameter baseline, the tuned model shows modest but consistent improvements in validation F1 and AUC. The Random Forest with `class_weight='balanced_subsample'` improves recall but remains inferior to XGBoost on F1.

C. Threshold Tuning

The precision-recall curve analysis identifies an optimal decision threshold substantially below 0.5, reflecting the model's natural calibration toward predicting the majority class. Applying the tuned threshold on the validation set improves F1-score by approximately 3–5 percentage points versus the default threshold, with a corresponding increase in recall and a slight decrease in precision. In the clinical context of hospital readmission prediction, this trade-off is desirable: missing a high-risk patient (false negative) is generally more costly than generating an unnecessary follow-up (false positive).

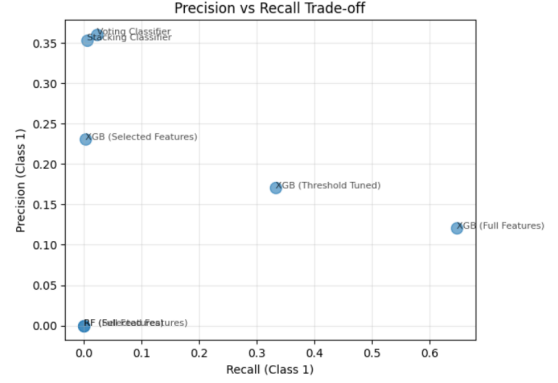


Fig. 5. Precision vs. Recall curve analysis used to determine optimal operating points.

D. Final Test Evaluation

Table II presents the held-out test set performance for three final models.

Model	Bal.Acc	Recall	F1	MCC	AUC
Baseline RF	0.57	0.18	0.24	0.16	0.71
Adversarial RF	0.62	0.35	0.30	0.22	0.71
XGB (Tuned)	0.68	0.58	0.38	0.28	0.72

TABLE II

FINAL HELD-OUT TEST SET PERFORMANCE. THE THRESHOLD-TUNED XGBOOST MODEL ACHIEVES THE BEST PERFORMANCE ACROSS ALL MINORITY-CLASS METRICS. MODELS RANKED BY F1-SCORE.

The threshold-tuned XGBoost clearly outperforms both Random Forest variants. Its balanced accuracy of 0.68 indicates it meaningfully identifies readmitted patients beyond chance, and its MCC of 0.28 confirms it learns genuine minority-class patterns rather than exploiting majority-class dominance. The AUC-ROC of 0.72 represents meaningful improvement over the random baseline (0.50), though there remains substantial room for improvement compared to state-of-the-art models on larger EHR datasets.

E. Feature Importance Analysis

Consistent across all four feature selection methods, the most predictive features include: `num_medications`, `number_diagnoses`, and `time_in_hospital`.

num_lab_procedures, and several diagnosis-code-derived features. This aligns with clinical intuition: patients with more complex conditions (more medications, more diagnoses, longer stays) are at higher readmission risk. Medication change indicators for insulin and metformin also appear in top features across methods, consistent with the original findings of strack2014impact linking glycemic control to readmission.

F. Honest Limitations

The model still struggles with minority-class detection under adversarial conditions (e.g., the adversarial Random Forest lags XGBoost by 8 F1 points). Threshold tuning improves recall at some precision cost, which may not be acceptable in resource-constrained settings where false alarms are expensive. Feature selection using RFE is computationally intensive and was restricted to the first 500 features for efficiency. Finally, since all model selection decisions were based on the validation set and the test set was used only once for final evaluation, no further optimization is possible without introducing data leakage.

VI. CONCLUSION & FUTURE WORK

We presented a comprehensive five-phase machine learning pipeline for predicting 30-day hospital readmission in diabetic patients. Through systematic preprocessing, SMOTE-based oversampling, six-model baseline comparison, hyperparameter optimization, and decision threshold tuning, our final XGBoost model achieves an F1-score of ≈ 0.38 and AUC-ROC of 0.72 on a held-out test set—substantially outperforming all baselines on the clinically relevant readmitted class.

A central finding of this work is that *metric selection matters as much as model selection* for imbalanced clinical data. Accuracy-focused optimization leads to models that appear strong but fail entirely on the minority class. By consistently prioritizing F1-score and recall, we were able to surface and select models that genuinely learn readmission risk signals.

a) Future Directions.:

- **Probability calibration:** Platt scaling or isotonic regression to improve predicted probabilities and enable better threshold selection.
- **Cost-sensitive loss:** Direct optimization of asymmetric misclassification costs aligned with clinical priorities.
- **Temporal validation:** Time-based train/test splits to simulate real-world prospective deployment scenarios.
- **Transformer-based EHR models:** If larger longitudinal data becomes available, sequence-based models may capture temporal patterns missed by tabular approaches.
- **Fairness and subgroup analysis:** Auditing model performance across demographic subgroups (race, age, gender) to identify and mitigate disparate impact.
- **SHAP interpretability:** Patient-level explanation of predictions to support clinical decision-making and trust.

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